

Explainable and Optimized Machine Learning Models for Diabetes Prediction

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Abstract—A key barrier to adopting machine learning (ML) in clinical environments is the lack of model interpretability, especially when applied to critical tasks such as early diabetes prediction. This study addresses the challenge by developing and evaluating five ML classifiers—Gradient Boosting, Support Vector Machine, Logistic Regression, Multi-Layer Perceptron, and Random Forest—using a dataset of 1,450 patient records from Local Hospital in Indonesia. After applying 5-fold cross-validation as a sampling technique to identify the best-performing fold, the models from this fold were then optimized through hyperparameter tuning and integrated into a soft voting classifier to enhance overall performance. The final predictions from the ensemble model were explained using SHAP and LIME to ensure transparency and interpretability. Experimental results demonstrate that the Voting Classifier achieved strong performance on the testing dataset, with an accuracy of 99.7%, precision of 100%, recall of 99.02%, F1-score of 99.51%, and an AUC of 0.9944. SHAP and LIME provided consistent and clinically meaningful explanations, with random blood glucose having the most dominant contribution in influencing the model output, followed by White Cell Count, Hemoglobin, Urea, Creatin serum identified as key predictors. These findings not only confirm the reliability of the model but also enhance its interpretability, making it suitable for clinical adoption. The explainability outputs from SHAP and LIME are well-suited for integration into predictive modeling dashboards to support real-time decision-making by medical professionals.

Keywords— diabetes prediction, machine learning, hyperparameter tuning, explainability methods

I. INTRODUCTION

Diabetes is one of the leading chronic diseases affecting millions globally, resulting in significant health complications and substantial economic burden. According to the IDF Diabetes Atlas 2021[1], around 537 million adults worldwide have diabetes, projected to reach 643 million by 2030 and 783 million by 2045. Global health spending on diabetes was USD 966 billion in 2021, a 316% increase over 15 years, showing its severe economic impact

Early and accurate prediction of diabetes is essential in preventing long-term complications and in improving overall healthcare outcomes. With the increasing availability of medical data, machine learning (ML) has become an effective tool for predictive modeling in healthcare. However, the adoption of ML in clinical settings is often limited by the lack of interpretability of the models, which is critical for building trust with healthcare professionals.

This research investigates the development and optimization of machine learning models for diabetes prediction, emphasizing both predictive performance and model interpretability. Specifically, we compare five classifiers: Gradient Boosting (GB), Support Vector Machine (SVM), Logistic Regression (LR), Multi-Layer Perceptron Classifier (MLP), and Random Forest (RF). We optimize their performance using hyperparameter tuning and employ Explainable Artificial Intelligence (XAI) techniques, namely SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), to make the predictions understandable.

The contributions of this study are threefold: (1) development and optimization of multiple ML models for diabetes prediction, (2) comparative analysis of model performance and interpretability using XAI methods, and (3) practical insights into feature importance that can assist medical professionals in decision-making.

II. RELATED WORK

Machine learning (ML) has emerged as a powerful tool for diabetes prediction, as demonstrated in several studies. Kavakiotis et al. [2] and Sisodia and Sisodia [3] highlight how ML models have achieved promising accuracy—up to 78%—in diagnosing diabetes from clinical data. To further enhance performance, Alghamdi et al. [4] and Li et al. [5] demonstrate the superiority of ensemble models such as Random Forest and XGBoost, particularly when combined with data balancing techniques and hyperparameter optimization.

In addition, Wicaksono et al. [6] compared Logistic Regression and Bayesian methods for predicting diabetes risk levels using clinical data from Indonesian patients, showing that Bayesian methods achieved the highest accuracy of 91%, while Logistic Regression obtained the highest ROC score of 0.988. It is important to note that the dataset used in this study is the same as the one utilized in Wicaksono et al. [6], ensuring consistency and comparability of experimental results. This indicates that traditional statistical learning models remain relevant and competitive when implemented with careful feature engineering and adequate data preparation.

However, while accuracy remains important, the growing demand for interpretable and transparent AI in healthcare has shifted attention toward Explainable Artificial Intelligence (XAI). Techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), introduced by Lundberg and Lee [7] and Ribeiro et al. [8] respectively, provide local and global insights into how ML models make predictions. SHAP, based

on cooperative game theory, offers consistent and theoretically grounded feature attributions with both local and global explanations, while LIME approximates complex models locally with interpretable models such as linear regressions, providing fast and intuitive explanations but with potential instability across perturbations. This difference suggests that SHAP is often preferred when global feature importance is critical, whereas LIME remains useful for quick local interpretability.

Raza Hasan et al. [9] further emphasize the importance of these methods in enhancing clinical trust and understanding of ML outcomes. In terms of model optimization, hyperparameter tuning is a critical step. Bergstra and Bengio [10] argue that methods like Random Search and Bayesian Optimization are more efficient than exhaustive Grid Search in identifying optimal model configurations. This optimization leads not only to improved accuracy but also better generalization to new patient data.

Equally important is feature selection, which helps reduce model complexity and enhance interpretability. Idris et al. [11] show that using techniques such as Recursive Feature Elimination (RFE) in conjunction with stacking and anomaly detection methods can reduce the number of features by half—without sacrificing model performance—while increasing the clinical relevance of the selected predictors. Recent studies further explore the integration of XAI with ensemble and optimized learning strategies. For example, Shou-Yu Lee et al. [12] combined SHAP with gradient boosting decision trees to offer accurate and interpretable clinical recommendations tailored to individual patients.

Collectively, these studies reflect a growing trend in diabetes prediction research: one that emphasizes not only accuracy and efficiency but also explainability and clinical usability. Building upon these insights, the present study proposes an integrated framework that applies ensemble-based ML models, optimized through hyperparameter tuning and refined via feature selection, while leveraging SHAP and LIME to ensure model decisions are transparent and interpretable in a clinical setting. It is important to note that the dataset used in this study is the same as the one utilized in Wicaksono et al. [6], ensuring consistency and comparability of experimental results.

III. METHODOLOGY

We used a supervised learning approach for binary classification of diabetes presence. The methodology consists of system design, data preprocessing, model training with hyperparameter tuning, and application of XAI techniques for interpretability.

A. System Design

The system is designed modularly in four main stages: data preprocessing, model training and tuning, model evaluation, explanation generation, and output to integrate dashboard predictive modeling. It begins with data cleaning and transformation, followed by the training of several machine learning models, evaluate the model, explainability techniques are then applied to make the model decisions interpretable.



Figure 1 System Design

B. Datasets

The dataset consists of 1,450 patient medical records collected from Local Hospital Balung, Jember, Indonesia, between September 26, 2014, and April 30, 2015. Attributes include gender, age, hemoglobin (g/dl), white cell count ($10^3/\mu\text{l}$), random blood glucose (mg/dl), serum creatinine (mg/dl), urea (mg/dl), total cholesterol (g/dl), and triglycerides (g/dl). The target variable indicates whether a patient has diabetes. This research was validated by Dr. Yudho S.Pd, a licensed medical doctor affiliated with the local hospital from which the data were obtained. Exploratory Data Analysis (EDA) was conducted to assess data characteristics. The class distribution was slightly imbalanced, with more non-diabetic (905 data) than diabetic cases (545 data).

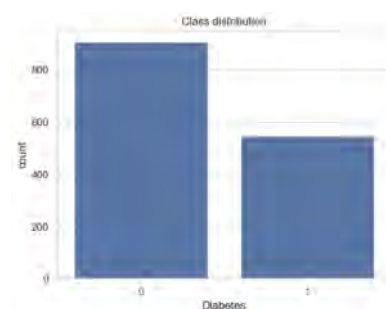


Figure 2 Class Distribution

The histograms in Fig. 3 shown provide a visual representation of the distribution of each numeric attribute in the dataset, which is a crucial step in the data preprocessing phase of machine learning. Attributes such as Age and Hemoglobin exhibit relatively symmetric, bell-shaped distributions, indicating a more balanced and potentially "normal" range of values. In contrast, variables like Random Blood Glucose, Creatinine Serum, and Urea are highly skewed to the right, meaning most values are concentrated on the lower end, with a few very high values that could be outliers. These types of distributions often require transformation (e.g., log scaling) to prevent models from being biased toward extreme values. Interestingly, Cholesterol shows a bimodal pattern, which might indicate the presence of distinct patient subgroups or conditions. Histograms revealed skewness in features like glucose and triglycerides, indicating potential outliers and clinical variability.

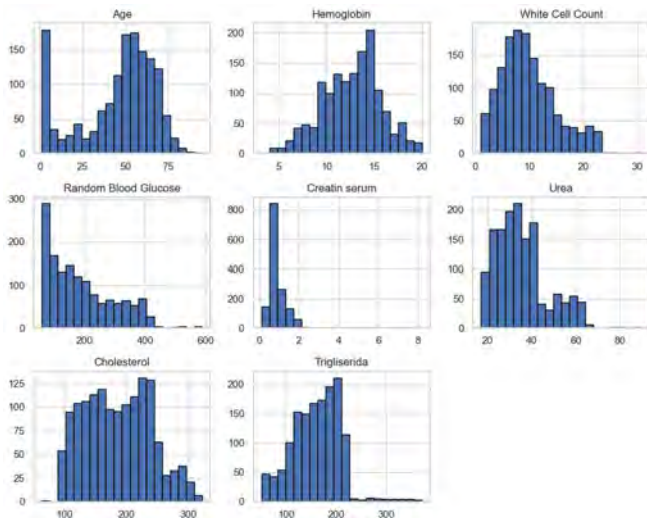


Figure 3 Numeric data Distribution

The correlation heatmap in Fig. 4 provides a visual overview of the linear relationships between different features in the dataset. In this chart, the color scale indicates the strength and direction of the correlations: red tones represent positive correlations, blue tones indicate negative correlations, and the intensity reflects the strength of the relationship.

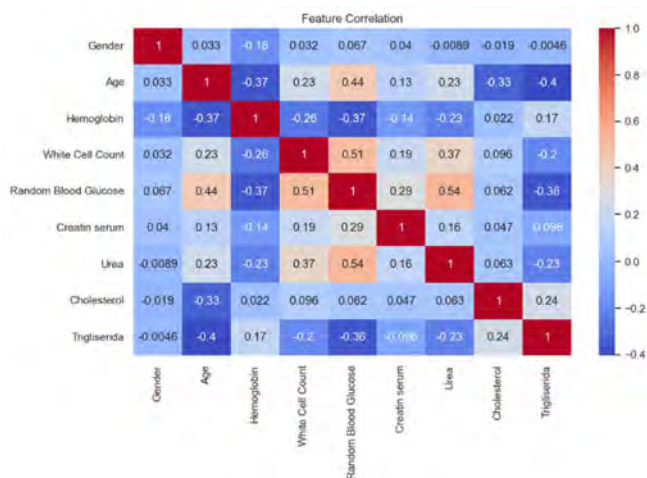


Figure 4 Feature Correlation

From the heatmap in Fig. 4, it can observe several notable patterns. White Cell count and Random Blood Glucose show a moderate positive correlation (0.51), suggesting that patients with higher random blood glucose tend to also have white cell count. Similarly, Urea is moderately correlated with Random Blood Glucose (0.54), which may imply a shared underlying medical condition affecting these values. Hemoglobin shows a moderate negative correlation with Random blood glucose (-0.37) and White Cell Count (-0.26), which might reflect anemia or inflammatory responses. Most other features, including Gender, show very weak or no significant correlation with others, indicating they may provide independent contributions to any predictive model.

C. Data Preprocessing

Missing values were handled using mean imputation for numerical features and mode imputation for categorical ones. The data was normalized using StandardScaler, and categorical variables were encoded numerically.

D. Model Training and Tuning

Five models were developed and tuned using GridSearchCV: Gradient Boosting, Support Vector Machine (SVM), Logistic Regression (LR), Multi-Layer Perceptron (MLP) Classifier, Random Forest. Initial parameter setting for each method uses parameters in table I.

Table I. Define Model with Hyperparameter grids

No.	Machine Learning Method	Initial Parameters
1	Gradient Boosting	'n_estimators': [100, 200], 'learning_rate': [0.05, 0.1], 'max_depth': [3, 5]
2	Support Vector Machine	'C': [1, 10], 'kernel': ['rbf', 'linear']
3	Logistic Regression	'C': [0.1, 1, 10]
4	MLP Classifier (Neural Network)	'hidden_layer_sizes': [(50, 100)], 'activation': ['relu', 'tanh']
5	Random Forest Classifier	'n_estimators': [100, 200], 'max_depth': [None, 10, 20], 'min_samples_split': [2, 5], 'min_samples_leaf': [1, 2]

Performance was evaluated based on accuracy, precision, recall, F1-score, and ROC AUC. The best parameters for each model were selected based on validation performance. A Voting Classifier using soft voting was also constructed to aggregate predictions from all models.

E. Voting Classifier

This study applies voting classifier to get the best model. There are two types of voting classifier: hard and soft. The difference between soft voting and hard voting in ensemble learning lies in the decision-making method of member models (base estimators).

- Hard Voting (Majority of votes)

In a hard voting classifier, each individual model provides a class prediction rather than a probability. The voting classifier then counts the frequency of each predicted class across all models. The class that receives the majority of votes is selected as the final prediction.

- Soft Voting (Average probability)

In a soft voting classifier, each model provides a predicted probability for each class using the

predict_proba() function. The probabilities from all models are then average for each class. The class with the highest average probability is selected as the final prediction.

This study uses a soft voting classifier with several considerations. While hard voting considers only the final predicted class of each model and selects the class with the majority of votes, it completely ignores how confident each model is about its predictions. In contrast, soft voting considers the probability scores (confidence levels) assigned by each model for each class. By averaging these probabilities, soft voting provides a more nuanced and informed decision.

F. Explainability

SHAP and LIME were used for model interpretability. SHAP provided both global and local explanations, while LIME offered instance-specific insights. These tools were applied to the best-performing models to explain feature contributions to predictions.

F.1. SHAP

The Shapley Additive Explanations (SHAP) method uses Shapley values to determine the contributions of features.

SHAP explains a model's output by computing the contribution of each feature to the difference between the actual prediction and the average prediction over all data.

The following is the SHAP formula for measuring the uncertainty of the feature importance distribution:

$$H(F) = -\sum_{i=1}^n p_i \log(p_i) \quad (1)$$

Where:

$$p_i = \frac{FI_i}{\sum_{j=1}^n FI_j} = \text{proportion of feature importance for feature } i$$

FI_i = feature importance value of feature i

n = total number of features

The lower the entropy, the more concentrated the feature importance is on a few features.

F.2. LIME

LIME (Local Interpretable Model-agnostic Explanations) is an interpretability technique that is model-agnostic, meaning it can be applied to any machine learning model, whether it's a black-box like Random Forests, SVMs, or Neural Networks.

Mathematically, LIME solves the following optimization problem:

$$\arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (2)$$

Where:

f = the original complex (black-box) model

g = the interpretable local surrogate model.

G = the class of interpretable models (e.g., linear models).

$\mathcal{L}(f, g, \pi_x)$ = a loss function that measures how closely the surrogate model g approximates the predictions of the original model f near the instance x , weighted by a proximity measure π_x

$\Omega(g)$ is a complexity penalty to ensure that the surrogate model remains simple and interpretable.

IV. EXPERIMENTS AND RESULTS

A. Performance Assessment with Baseline Parameters

Experiments were initially conducted using 5-fold cross-validation using baseline parameter to evaluate model performance across different data splits. Cross-validation is considered the gold standard in machine learning as it maintains the fairness of the methods applied. The performance results obtained for each fold across all methods used are presented in Table II.

Table II. Area under ROC curve (AUC) Hyperparameter tuning and Performance Metrics of the Best Model

Fold	Gradient bosting	Support Vector Machine	Logistic Regression	Multi-Layer Perceptron	Random Forest
1	0.9888	0.9903	0.9916	0.9906	0.9906
2	0.9898	0.9881	0.9888	0.9909	0.9979
3	0.9945	0.9989	0.9985	0.9999	0.9999
4	1.0000	0.9999	0.9987	0.9996	1.0000
5	0.9818	0.9974	0.9886	0.9934	0.9965
Average ROC AUC per fold (across models):					
Fold 1: 0.9901			Fold 4: 0.9997		
Fold 2: 0.9911			Fold 5: 0.9915		
Fold 3: 0.9983					

According to the results shown in the Table II, the best performance was achieved in fold 4.

B. Performance Metrics of Voting Classifier

The subsequent process involved conducting hyperparameter tuning using the training data from fold 4, followed by implementing the voting classifier, which was then applied to the testing data. Before applying the Voting Classifier, the results of the hyperparameter tuning of the five machine learning methods are shown in Table III below.

Table III. Hyperparameter tuning and Performance Metrics of the Best Model

N ^o	Machine learning method	Best parameters	Accuracy	Precision	Recall	F1-score	AUC
1	Gradient bosting	{'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200}	99.31%	99.02%	99.02%	99.02%	0.99
2	Support Vector Machine	{'C': 1, 'kernel': 'rbf'}	98.97%	99.01%	98.04%	98.52%	1.00
3	Logistic Regression	{'C': 10}	99.31%	99.02%	99.02%	99.02%	1.00
4	Multi-Layer Perceptron	{'activation': 'relu', 'hidden_layer_sizes': (50,)}	99.31%	99.02%	99.02%	99.02%	0.99
5	Random Forest	{'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}	99.31%	99.02%	99.02%	99.02%	0.99

And after the voting classifier was conducted on 5 (five) machine learning methods, it can be seen that the Voting Classifier demonstrated competitive performance, combining strengths from all models.

Voting Classifier achieved strong performance on the testing dataset, with an accuracy of 99.7%, precision of 100%, recall of 99.02%, F1-score of 99.51%, and an AUC of 0.9944.

Fig 5 is the result of confusion matrix from Voting classifier model applied to testing data. Only 1 (one) case is false negative.

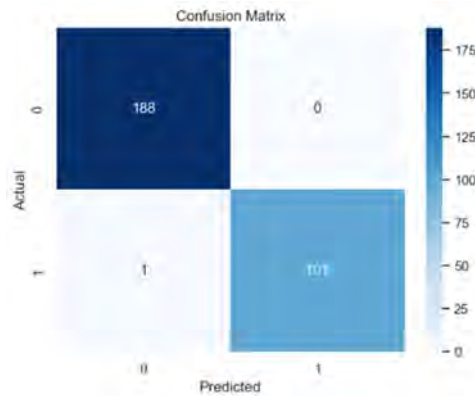


Figure 5 Confusion Matrix of Voting Classifier of testing dataset

Meanwhile, the area under the ROC curve of the voting classifier applied to the testing data is in Fig. 6.

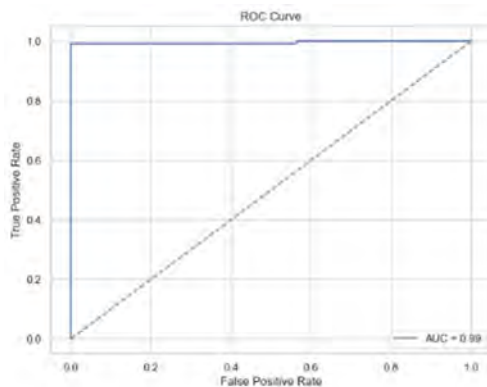


Figure 6 ROC Curve of Voting Classifier on testing dataset

C. Explainability Insights

B.1. SHAP

SHAP highlighted the following as the most influential features: random blood glucose, white cell count, hemoglobin, urea and creatin serum. From Fig 7, SHAP value reflects how much influence a feature has on the model output. Red means high feature value and blue means low feature value.

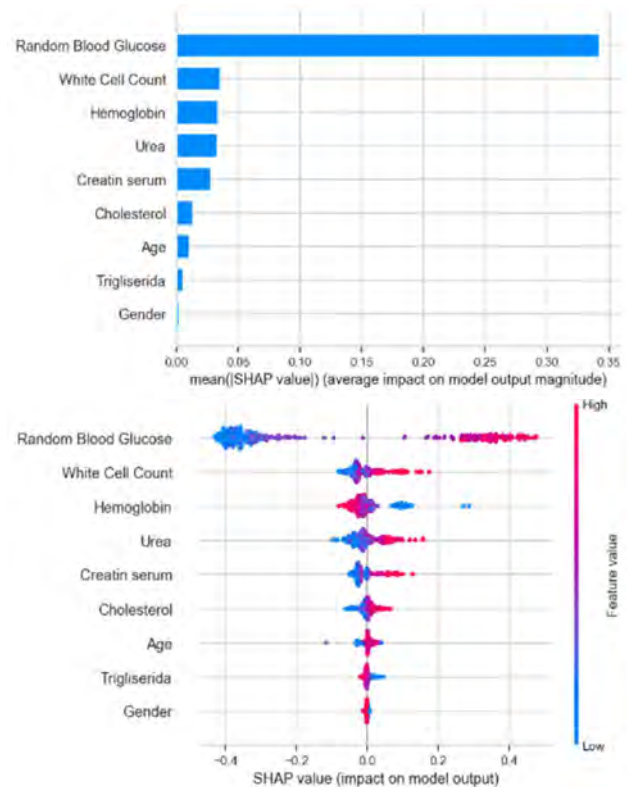


Figure 7 SHAP summary and bar plot

Random Blood Glucose has the strongest impact, where high levels push predictions toward disease and low levels reduce risk. White Cell Count follows, with higher values slightly increasing the chance of a positive prediction. Hemoglobin shows that lower levels tend to increase disease risk, while higher levels reduce it. Urea contributes to higher disease probability when elevated. Similarly, higher Creatin Serum levels slightly raise disease predictions. Other features has relatively low impact.

B.2. LIME

The following is an example of a LIME explanation for one patient's data.



Figure 8 LIME Local Explanation Chart

LIME provided consistent local explanations across individual predictions, supporting the transparency and reliability of model outputs. The horizontal axis shows the direction of feature contribution:

- Left: supports the prediction of "No Diabetes".
- Right: supports the prediction of "Diabetes".

Accompanied by the actual feature values on the right side. Based on the LIME diagram in Fig 8, the model prediction is: no diabetes. Features such as Random Blood Glucose = 155.00, Urea = 20.00, White Cell Count = 4.40, and others contribute towards "No Diabetes" (blue bar to the left).

V. CONCLUSION

This study presented a comparative analysis of optimized machine learning models for diabetes prediction, emphasizing both predictive performance evaluation and interpretability. Initially, 5-fold cross-validation was conducted to select the fold yielding optimal performance. The models from this selected fold were then subjected to hyperparameter tuning before being integrated into a soft voting classifier to maximize overall model performance.

By integrating multiple models, the Voting Classifier delivered competitive accuracy and robust performance. Voting Classifier achieved strong performance on the testing dataset, with an accuracy of 99.7%, precision of 100%, recall of 99.02%, F1-score of 99.51%, and an AUC of 0.9944. When combined with SHAP and LIME, provided meaningful insights into model predictions. These insights can aid clinicians in understanding and trusting model outputs.

Explainable AI (XAI) methods implemented in this research can also be integrated into a disease prediction dashboard, enabling medical staff to interpret and validate the predictions generated by machine learning models. This integration has the potential to enhance clinical decision-making and foster greater trust in AI-driven diagnostics.

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